SOFTWARE REQUIREMENTS SPECIFICATION

Digital Directory System

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines all functional and non-functional requirements for the Digital Directory platform — a multi-country, multi-city business directory system built on a single centralized backend with country-based subdomain routing. The document is intended to serve as the complete technical and operational blueprint for design, development, testing, and client presentation.

1.2 Scope

Digital Directory is a subscription-based, scalable web platform that enables business owners across multiple countries to register, manage, and promote their businesses. The system supports multi-country and multi-city structures, business profiles with media, product and service listings, branch management, subscription plans, payment processing, and a full administrative control panel. All operations are managed through one backend system with intelligent subdomain-based country routing.

1.3 Intended Audience

This document is prepared for the project client, the backend development team at Labverse.org, and any future stakeholders or reviewers involved in the delivery or maintenance of the Digital Directory platform.

1.4 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
Subdomain	A prefix to the main domain that identifies a country, e.g. uk.digitaldirectory.com
Business Owner	A registered user who creates and manages one or more business listings
Super Admin	The top-level administrator with full system access
JWT	JSON Web Token — used for stateless authentication
Stripe	Third-party payment processing provider
NodeMailer	Node.js library used to send transactional emails
Cloudflare R2	Object storage for media files, integrated with Cloudflare CDN
RBAC	Role-Based Access Control

Term	Definition
FK / PK	Foreign Key / Primary Key — relational database constraints
CDN	Content Delivery Network for fast static asset delivery
Webhook	HTTP callback triggered automatically by Stripe on payment events

2. System Overview

Digital Directory is a centralized multi-tenant platform where a single backend serves all countries simultaneously. Country filtering is handled dynamically through subdomain extraction from the HTTP request, ensuring that each visitor automatically sees only the businesses relevant to their country without any manual selection required.

The architecture is designed around the principle of a single source of truth — one backend, one database, and one unified admin panel — while maintaining logical separation of country-specific data through intelligent query filtering. This approach eliminates infrastructure redundancy, simplifies maintenance, and enables unlimited geographic expansion without additional servers.

2.1 High-Level Architecture

Layer	Technology	Responsibility
DNS & CDN	vesel	Wildcard subdomain routing, SSL termination, media CDN
Frontend	Web Application	Country-specific UI served per subdomain
Backend	Node.js + Express	Single API handling all countries via subdomain context
Database	PostgreSQL	Centralized data store with relational integrity
Media Storage	Cloudflare R2	Scalable object storage for images, videos, PDFs
Authentication	JWT (JSON Web Tokens)	Stateless, role-based auth across all users
Payments	Stripe	Subscription billing, webhooks, invoicing
Email	NodeMailer	Transactional notifications and alerts

2.2 Subdomain Examples

The platform uses wildcard DNS to route country-specific traffic. Every subdomain points to the same server, and the backend determines the active country from the request host header.

Country	Subdomain URL	Country Code
United Kingdom	uk.digitaldirectory.com	UK
South Korea	korea.digitaldirectory.com	KR
Pakistan	pakistan.digitaldirectory.com	PK

United States	us.digitaldirectory.com	US
Germany	germany.digitaldirectory.com	DE
Any New Country	country.digitaldirectory.com	Auto-configured

3. Core Modules

The platform is composed of eight major functional modules, each designed to operate independently while contributing to the unified system. Together they form the complete Digital Directory ecosystem.

3.1 Module 1 — Admin Panel

The Admin Panel is the centralized command center for the entire platform. The Super Admin has unrestricted access to all system functions across every country and city.

- **Country Management:** Create, edit, and deactivate countries. Each country record includes its name, country code, and unique subdomain identifier.
- **City Management:** Add and manage unlimited cities under each country. Cities are always linked to their parent country.
- **User Management:** View all registered users, suspend or deactivate accounts, and reset passwords on behalf of users.
- **Business Listing Management:** Review submitted business profiles, approve or reject listings, and manage active or suspended listings.
- **Subscription Package Management:** Create, edit, and deactivate subscription plans with configurable pricing and duration.
- **Payment Monitoring:** View all payment transactions, subscription statuses, and revenue metrics by country.
- **Notification Broadcasting:** Send system-wide or country-specific announcements to all users via email.
- **Analytics Dashboard:** Access high-level metrics including active businesses per country, subscription conversion rates, and payment volumes.
- **System Logs:** View activity logs for auditing, debugging, and compliance purposes.

3.2 Module 2 — Multi-Country and Multi-City Support

The platform is built to support an unlimited number of countries and cities without any architectural changes. Adding a new country requires only a database entry and a DNS record — no new server or codebase is needed.

- Unlimited countries can be registered and activated by the admin.
- Each country supports unlimited cities.
- Country context is automatically inferred from the subdomain on every API request.
- All database queries are automatically scoped to the active country, preventing data bleed between regions.

3.3 Module 3 — Business Registration and Profile Management

Business owners can register on the platform, create detailed business profiles, and manage all aspects of their digital presence. A single user account may manage multiple businesses across different cities.

- **Business Details:** Business name, description, industry category, phone number, email address, website URL, and physical address.
- **Products and Services:** Business owners can add, edit, and remove individual products or services with descriptions, pricing, and images.
- **Branch Management:** A business can register multiple physical branches, each with its own city, address, phone, and operating hours.
- **Media Uploads:** Owners can upload a logo, multiple gallery images, video links or video files, and a business card in image or PDF format.
- **Business Card:** A dedicated business card asset (image or PDF) can be uploaded per business for brand identity display.
- **Profile Editing:** All profile fields, media, and branch information can be updated at any time from the owner dashboard.
- **Listing Visibility:** A listing becomes publicly visible only after admin approval and an active subscription. Expired subscriptions automatically deactivate the listing.

3.4 Module 4 — Subscription and Payment System

All business listings are gated behind a subscription plan. The payment infrastructure is built on Stripe, ensuring secure, PCI-compliant transactions globally.

- **Monthly Plan:** Business owners pay a monthly fee to maintain an active listing. Auto-renewal is supported.
- **Yearly Plan:** A discounted annual subscription is available. Auto-renewal is supported.
- **Secure Checkout:** Payment is processed through the Stripe-hosted checkout flow with full SSL encryption.
- **Webhook Integration:** Stripe sends real-time event notifications to the backend on payment success, failure, and subscription renewals. The backend updates subscription statuses automatically.
- **Invoice Generation:** A payment receipt and invoice are automatically generated and emailed to the business owner after every successful transaction.
- **Subscription Expiry Logic:** The system checks subscription end dates daily. Listings with expired subscriptions are automatically deactivated and the owner is notified.

3.5 Module 5 — Notification System

The notification system is built on NodeMailer and handles all transactional communication between the platform and its users. Notifications are triggered automatically by system events and can also be sent manually by the admin.

Notification Event	Recipient	Trigger
Account Registration	Business Owner	On successful email verification
Listing Submitted	Admin	When a business submits for approval
Listing Approved	Business Owner	When admin approves the listing
Listing Rejected	Business Owner	When admin rejects the listing with reason
Payment Confirmed	Business Owner	On successful Stripe payment
Payment Failed	Business Owner	On failed or declined payment
Subscription Expiry (7d)	Business Owner	Seven days before subscription ends
Subscription Expiry (1d)	Business Owner	One day before subscription ends
Subscription Expired	Business Owner	On day of expiry
Password Reset	Business Owner	On password reset request
Admin Broadcast	All / Country	On manual admin send

3.6 Module 6 — Role-Based Access Control (RBAC)

The system enforces strict role-based access control across all routes and actions. Each role has a clearly defined set of permissions that cannot be escalated by the user.

Role	Access Level	Key Permissions
Super Admin	Full System	All CRUD operations, payments, users, countries, analytics, notifications
Business Owner	Own Business Only	Create and manage own business, upload media, manage subscription, view invoices
Public User	Read-Only	Browse businesses, search and filter, view profiles — no account required

3.7 Module 7 — Search and Filtering

The public directory interface provides a comprehensive search and filter system that allows visitors to quickly find relevant businesses within their country.

- **Country Filter:** Automatically applied based on the active subdomain. Cannot be overridden by the public user.
- **City Filter:** Dropdown selection of all cities within the active country.
- **Category Filter:** Filter businesses by their assigned industry or service category.
- **Keyword Search:** Full-text search across business name, description, and service listings.

- **Product or Service Type:** Filter by specific product or service tags associated with business listings.

3.8 Module 8 — Business Branch Management

To accommodate businesses with multiple physical locations, the platform supports branch creation as a first-class feature within each business profile.

- Each business can have one or more branches in addition to the primary address.
- Each branch stores its own city, street address, phone number, and operating hours.
- Branches are displayed on the public business profile as expandable location entries.
- Branches are managed entirely from the business owner's dashboard.

4. User Journeys

This section describes the complete end-to-end experience for each user role within the system. These journeys reflect the full operational flow from onboarding through active usage.

4.1 Public User Journey

A public visitor arrives on a country-specific subdomain and can immediately browse, search, and explore business listings without needing an account.

Step 1 **Visit Subdomain —** User navigates to a country URL such as uk.digitaldirectory.com. The backend detects the subdomain and scopes all data to the United Kingdom.

Step 2 **Browse Listings —** The homepage displays approved, active business listings from the active country, organized by category or recent activity.

Step 3 **Search and Filter —** The visitor uses keyword search, city dropdown, and category filter to narrow results to their needs.

Step 4 **View Business Profile —** Clicking a listing opens the full business profile with description, media gallery, services, branches, and contact information.

Step 5 **Contact Business —** The visitor uses the displayed phone number, email, or website link to contact the business directly. No account is required.

4.2 Business Owner Journey — Registration and Onboarding

A new business owner registers on the platform, builds their business profile, selects a subscription, and awaits admin approval before going live.

Step 1 **Registration —** Owner visits the platform and submits name, email, and password. A verification email is sent.

Step 2 **Email Verification —** Owner clicks the link in the verification email to activate their account.

Step 3 **Login —** Owner logs in with their verified credentials. A JWT token is issued.

Step 4 **Create Business —** Owner fills in business name, description, category, city, phone, email, website, and address.

Step 5 **Upload Media —** Owner uploads logo, gallery images, video link or file, and a business card image or PDF.

Step 6 **Add Products/Services —** Owner lists the products or services offered, with descriptions and optional pricing.

Step 7 **Add Branches —** Owner adds one or more branch locations with individual city, address, and contact details.

Step 8 **Select Subscription —** Owner chooses either the monthly or yearly subscription plan.

Step 9 **Stripe Payment —** Owner completes payment on the secure Stripe checkout. A payment confirmation email is sent.

Step 10 **Admin Review —** The submitted listing enters the admin review queue. The admin approves or rejects with feedback.

Step 11 **Business Published —** On approval, the listing becomes publicly visible on the correct country subdomain.

4.3 Business Owner Journey — Subscription Renewal

When an active subscription approaches expiry, the system guides the owner through a seamless renewal process.

Step 1 **Expiry Reminder —** The system automatically sends reminder emails 7 days and 1 day before the subscription end date.

Step 2 **Login to Dashboard —** Owner logs in and sees a renewal prompt on their dashboard.

Step 3 **Select Plan —** Owner selects the same or a different subscription plan for renewal.

Step 4 **Complete Payment —** Owner completes the Stripe payment. The webhook confirms the transaction to the backend.

Step 5 **Subscription Updated —** The subscription end date is extended and the listing remains active without interruption.

4.4 Admin Journey — Daily Operations

The admin manages all platform activity through a centralized dashboard covering listings, payments, geography, and communications.

Step 1 **Admin Login —** Admin authenticates with elevated credentials. A JWT token with admin role is issued.

Step 2 **Dashboard Overview —** Admin sees a summary of pending approvals, active businesses, recent payments, and system alerts.

Step 3 **Manage Geography —** Admin adds or updates countries and cities as the platform expands to new regions.

Step 4 **Review Submissions —** Admin opens the listing approval queue, reviews each business profile, and approves or rejects with optional reason.

Step 5 **Monitor Payments —** Admin views all transactions, filters by country or date range, and checks subscription statuses.

Step 6 **Send Notifications —** Admin composes and dispatches broadcast emails to all users or users within a specific country.

Step 7 **View Reports —** Admin accesses analytics showing business counts, subscription rates, and revenue by country.

5. Subdomain-Based Country Architecture

The subdomain routing mechanism is the architectural cornerstone of the entire platform. It enables a single backend and database to serve multiple countries intelligently without any duplication of infrastructure.

5.1 DNS Configuration

Versel manages the DNS for the root domain digitaldirectory.com. A wildcard DNS record is configured to point all subdomains to the same server IP address. This means whether a user visits uk.digitaldirectory.com or korea.digitaldirectory.com, the request arrives at the same backend server. SSL certificates are also issued automatically by Versel for all subdomains under the wildcard.

5.2 Backend Country Resolution Logic

When the backend receives an incoming request, it follows a deterministic five-step process to resolve the country context. This logic is applied as middleware on every API request, ensuring all downstream queries are automatically scoped to the correct country.

Step 1 **Read Host Header —** The backend reads the incoming HTTP request's host header value.

Step 2 **Extract Subdomain —** The subdomain prefix is extracted from the host. For uk.digitaldirectory.com, the extracted value is 'uk'.

Step 3 **Validate Against Database —** The extracted subdomain is matched against the countries table. If no match is found, a 404 response is returned.

Step 4 **Attach Country Context —** The resolved country record, including its ID and country code, is attached to the request object for use by all subsequent route handlers.

Step 5 **Scope All Queries —** Every database query within the request lifecycle automatically includes a filter on country_id, ensuring only data belonging to the resolved country is returned.

5.3 Benefits of This Architecture

- No separate backend is needed per country — a single deployment serves all regions.
- Expanding to a new country requires only a new database record and a DNS entry.
- All security, validation, and business logic is maintained in a single codebase.
- Data isolation between countries is enforced at the query level — not at the infrastructure level.
- The architecture supports horizontal scaling since the backend is entirely stateless.

6. Functional Requirements

This section defines all functional capabilities the system must fulfill. Each requirement is presented in terms of what the system does in response to user or administrative actions.

6.1 User Management

- The system must allow new users to register with a name, email address, and password.
- The system must send an email verification link upon registration and disallow login until the email is verified.
- The system must authenticate users using JWT-based stateless sessions with configurable expiry.
- The system must allow users to request a password reset via a time-limited email link.
- The system must allow users to update their profile information from their dashboard.
- The admin must be able to suspend, deactivate, or permanently delete any user account.

6.2 Business Management

- The system must allow an authenticated business owner to create one or more business listings.
- Each business must be associated with a country, city, and category at creation time.
- The system must allow owners to add, edit, and remove products or services linked to their business.
- The system must allow owners to create and manage branch locations, each with its own address and contact details.
- The system must allow owners to upload a logo, up to a configurable number of gallery images, one business card asset, and a video link or file.
- The system must enforce that a business listing is only publicly visible if it has an active subscription and has been approved by the admin.
- The system must allow owners to deactivate their own listing at any time.

6.3 Admin Controls

- The admin must be able to create, update, and deactivate country records including subdomain and country code.
- The admin must be able to create, update, and remove city records linked to a country.
- The admin must be able to create and manage industry categories shared across all countries.
- The admin must be able to approve or reject any business listing submission, with an optional rejection reason communicated to the owner.
- The admin must be able to suspend or reinstate any user account.
- The admin must be able to create, edit, and deactivate subscription plans with custom pricing and duration.

- The admin must be able to view full system logs including user actions, payment events, and listing changes.
- The admin must be able to send email notifications to all users, users within a specific country, or individual accounts.

6.4 Subscription and Payment

- The system must integrate with Stripe to process monthly and yearly subscription payments.
- The system must listen to Stripe webhook events and update subscription statuses in real time on payment success or failure.
- The system must generate and email an invoice to the business owner on every successful payment.
- The system must automatically check subscription expiry dates and deactivate expired listings.
- The system must send reminder emails 7 days and 1 day before a subscription expires.
- The system must support auto-renewal when enabled by the business owner via Stripe recurring billing.

6.5 Notifications

- All notification emails must be sent via NodeMailer with HTML templates.
- Email triggers must include: registration, approval, rejection, payment confirmation, payment failure, subscription expiry (7 days), subscription expiry (1 day), subscription expired, and password reset.
- The admin broadcast function must support sending emails to all users globally or filtered by country.
- Failed email deliveries must be logged for admin review and retry.

6.6 Search and Discovery

- The system must automatically apply country context to all search results based on the active subdomain.
- The system must support filtering by city within the active country.
- The system must support filtering by business category.
- The system must support full-text keyword search across business name, description, and associated services.
- The system must return only approved, active-subscription businesses in all public search results.

7. Non-Functional Requirements

7.1 Performance

- All public-facing pages and API responses must load within three seconds under normal traffic conditions.
- Database queries must be indexed on all foreign keys and commonly filtered fields including country_id, city_id, category_id, is_approved, and subscription status.
- Media assets must be served through Cloudflare R2 with CDN caching to minimize latency across geographies.
- The backend must be capable of handling at least 500 concurrent users per country subdomain without degradation.

7.2 Security

- All traffic must be served over HTTPS with SSL certificates managed by Cloudflare.
- All user passwords must be hashed using bcrypt with a minimum of 12 salt rounds before storage.
- JWT tokens must include role claims and have a defined expiry. Refresh tokens must be supported.
- All API endpoints must enforce role-based authorization, rejecting requests where the caller's role does not permit the action.
- All user inputs must be validated and sanitized server-side to prevent SQL injection, XSS, and other injection attacks.
- API rate limiting must be enforced on authentication endpoints and public search routes to prevent abuse.
- Stripe webhook payloads must be verified using Stripe's signing secret before processing.

7.3 Scalability

- The backend must be stateless so that additional server instances can be added horizontally without session conflict.
- Media storage must be offloaded entirely to Cloudflare R2, removing local disk dependency and enabling unlimited media scale.
- Database design must support adding new countries without schema changes — only new row insertions.
- The platform must support containerized deployment to facilitate auto-scaling in cloud environments.

7.4 Availability and Reliability

- The platform must target 99% uptime availability on an annual basis.
- Automated database backups must run daily with a minimum 30-day retention policy.
- Critical background jobs including subscription expiry checks and webhook processing must include retry logic for failure recovery.
- System health checks and uptime monitoring must be configured and alert the admin team on downtime.

7.5 Maintainability

- The backend codebase must follow RESTful API design conventions with clear endpoint naming.
- All environment-sensitive configuration including API keys, database credentials, and Stripe secrets must be stored in environment variables, never in code.
- Database migrations must be version-controlled using a migration tool to allow safe schema evolution.
- System activity logs must be retained for a minimum of 90 days and be accessible to the admin panel.

8. Database Design

The database is designed to be normalized, non-redundant, and optimized for the country-scoped query patterns of the platform. Every table follows clear relational integrity rules, and indexes are placed on all fields used in filtering, joining, and lookup operations. The design supports unlimited growth across countries, cities, businesses, and subscriptions without structural changes.

8.1 users

Stores all registered user accounts regardless of role. The role field distinguishes between super admins and business owners.

Column	Type	Constraints	Description
id	UUID	Primary Key	Globally unique user identifier
name	VARCHAR(150)	NOT NULL	Full display name of the user
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email address
password_hash	TEXT	NOT NULL	bcrypt-hashed password
role	ENUM	NOT NULL, DEFAULT 'owner'	Values: super_admin, owner
is_verified	BOOLEAN	DEFAULT false	Email verification flag
is_active	BOOLEAN	DEFAULT true	Admin suspension flag
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp
updated_at	TIMESTAMP	Auto-updated	Last profile update timestamp

8.2 countries

Defines each supported country. The subdomain field is used to match incoming request host headers.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique country identifier
name	VARCHAR(100)	NOT NULL	Country display name
country_code	VARCHAR(10)	UNIQUE, NOT NULL	ISO code, e.g. UK, KR, PK
subdomain	VARCHAR(50)	UNIQUE, NOT NULL	Subdomain prefix, e.g. 'uk', 'korea'

Column	Type	Constraints	Description
is_active	BOOLEAN	DEFAULT true	Controls whether the country subdomain is live

8.3 cities

Stores cities linked to their parent country. Used for listing assignment and search filtering.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique city identifier
name	VARCHAR(100)	NOT NULL	City display name
country_id	UUID	Foreign Key → countries	Parent country reference
is_active	BOOLEAN	DEFAULT true	Visibility flag

8.4 categories

Shared taxonomy of business categories, used for filtering and organizing listings across all countries.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique category identifier
name	VARCHAR(100)	UNIQUE, NOT NULL	Category label, e.g. Restaurants, Legal, Healthcare
is_active	BOOLEAN	DEFAULT true	Soft-delete flag

8.5 businesses

The central entity of the platform. Each business belongs to a user, a country, a city, and a category. Approval and subscription status control public visibility.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique business identifier
user_id	UUID	Foreign Key → users	Owner of the business

Column	Type	Constraints	Description
country_id	UUID	Foreign Key → countries	Country the listing belongs to
city_id	UUID	Foreign Key → cities	Primary city of the business
category_id	UUID	Foreign Key → categories	Industry category
name	VARCHAR(200)	NOT NULL	Business display name
description	TEXT	NULLABLE	Full business description
phone	VARCHAR(30)	NULLABLE	Primary contact phone
email	VARCHAR(255)	NULLABLE	Business contact email
website	VARCHAR(300)	NULLABLE	Business website URL
address	TEXT	NULLABLE	Primary physical address
logo_url	TEXT	NULLABLE	Cloudflare R2 URL for logo
is_approved	BOOLEAN	DEFAULT false	Admin approval flag
is_active	BOOLEAN	DEFAULT true	Owner-controlled visibility flag
created_at	TIMESTAMP	DEFAULT NOW()	Submission timestamp
updated_at	TIMESTAMP	Auto-updated	Last edit timestamp

8.6 business_branches

Stores additional physical locations for a business. Each branch is tied to a city within the same country as the parent business.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique branch identifier
business_id	UUID	Foreign Key → businesses	Parent business
city_id	UUID	Foreign Key → cities	City of this branch
address	TEXT	NOT NULL	Branch street address
phone	VARCHAR(30)	NULLABLE	Branch contact phone
operating_hours	TEXT	NULLABLE	Open hours as text or JSON
created_at	TIMESTAMP	DEFAULT NOW()	Branch creation timestamp

8.7 business_services

Stores individual products or services offered by a business. Supports detailed listing enrichment and service-type filtering.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique service identifier
business_id	UUID	Foreign Key → businesses	Parent business
title	VARCHAR(200)	NOT NULL	Product or service name
description	TEXT	NULLABLE	Detailed description
price	DECIMAL(10, 2)	NULLABLE	Optional display price
image_url	TEXT	NULLABLE	Associated image on Cloudflare R2
created_at	TIMESTAMP	DEFAULT NOW()	Creation timestamp

8.8 business_media

Stores all gallery images and video assets for a business. The media_type field distinguishes between image and video entries.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique media item identifier
business_id	UUID	Foreign Key → businesses	Parent business
media_type	ENUM	NOT NULL	Values: image, video
media_url	TEXT	NOT NULL	Cloudflare R2 URL or external video URL
sort_order	INT	DEFAULT 0	Display order for gallery
created_at	TIMESTAMP	DEFAULT NOW()	Upload timestamp

8.9 business_cards

Stores the business card asset for each business. A business can have one active business card, either as an image or a PDF file.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique card identifier
business_id	UUID	UNIQUE, FK → businesses	Parent business — one card per business
card_url	TEXT	NOT NULL	Cloudflare R2 URL for the card file
file_type	ENUM	NOT NULL	Values: image, pdf
created_at	TIMESTAMP	DEFAULT NOW()	Upload timestamp

8.10 subscription_plans

Defines the available subscription tiers. Plans are managed by the admin and can be updated or retired without deleting existing subscriber records.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique plan identifier
name	VARCHAR(100)	NOT NULL	Plan label, e.g. Monthly, Yearly
price	DECIMAL(10, 2)	NOT NULL	Billing amount in base currency
duration_in_days	INT	NOT NULL	Subscription length in days
is_active	BOOLEAN	DEFAULT true	Controls whether this plan is available for purchase
stripe_price_id	VARCHAR(100)	NULLABLE	Stripe Price ID for recurring billing

8.11 subscriptions

Tracks the subscription status of each business. A business can only have one active subscription at a time. Historical records are retained for audit.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique subscription record

Column	Type	Constraints	Description
business_id	UUID	Foreign Key → businesses	Subscribed business
plan_id	UUID	Foreign Key → subscription_plans	Plan at time of purchase
stripe_subscription_id	VARCHAR(100)	NULLABLE	Stripe subscription object ID
start_date	DATE	NOT NULL	Subscription start date
end_date	DATE	NOT NULL	Subscription expiry date
status	ENUM	NOT NULL	Values: active, expired, cancelled, pending
auto_renew	BOOLEAN	DEFAULT false	Auto-renewal flag
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp

8.12 payments

Records every payment transaction linked to a subscription. Provides a complete financial audit trail.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique payment record
subscription_id	UUID	Foreign Key → subscriptions	Linked subscription
stripe_payment_intent	VARCHAR(100)	UNIQUE, NOT NULL	Stripe PaymentIntent ID
amount	DECIMAL(10,2)	NOT NULL	Amount charged in base currency
currency	VARCHAR(10)	NOT NULL, DEFAULT 'usd'	ISO currency code
status	ENUM	NOT NULL	Values: success, failed, refunded
created_at	TIMESTAMP	DEFAULT NOW()	Payment event timestamp

8.13 notifications_log

Records all outbound notifications for auditing and retry management. This table enables the admin to monitor delivery failures and track communication history.

Column	Type	Constraints	Description
id	UUID	Primary Key	Unique notification record
user_id	UUID	FK → users	Recipient user
type	VARCHAR(50)	NOT NULL	Notification type, e.g. approval, payment, expiry
channel	ENUM	NOT NULL	Values: email (sms in future)
status	ENUM	NOT NULL	Values: sent, failed, pending
sent_at	TIMESTAMP	NULLABLE	Delivery timestamp
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp

8.14 Entity Relationship Summary

The following describes the primary relationships between all tables in plain terms.

- **users to businesses:** One user can own many businesses (one-to-many).
- **businesses to countries:** Each business belongs to exactly one country (many-to-one).
- **businesses to cities:** Each business belongs to exactly one primary city (many-to-one).
- **businesses to categories:** Each business belongs to exactly one category (many-to-one).
- **businesses to business_branches:** One business can have many branches (one-to-many).
- **businesses to business_services:** One business can have many services or products (one-to-many).
- **businesses to business_media:** One business can have many media items (one-to-many).
- **businesses to business_cards:** One business has at most one business card (one-to-one).
- **businesses to subscriptions:** One business can have many subscription records over time (one-to-many), with one active at any time.
- **subscriptions to payments:** One subscription can have many payment records (one-to-many).
- **cities to countries:** Each city belongs to exactly one country (many-to-one).

9. Technology Stack

The following technology choices have been made to ensure performance, scalability, security, and long-term maintainability of the Digital Directory platform.

Layer	Technology	Rationale
Backend Runtime	Node.js	High-performance, event-driven, ideal for I/O-heavy API workloads
Web Framework	Express.js	Minimal, flexible, widely supported REST API framework
Database	PostgreSQL	Relational integrity, strong indexing, JSON support, and proven scalability
Authentication	JWT	Stateless tokens enable horizontal scaling without shared session state
Media Storage	Cloudflare R2	S3-compatible object storage with zero egress fees and global CDN integration
DNS and SSL	versel	Wildcard DNS, automatic SSL for all subdomains, DDoS protection
Payment Gateway	Stripe	Global reach, subscription billing, webhooks, PCI compliance built-in
Email Notifications	NodeMailer	Flexible SMTP integration for transactional emails with HTML templates
Password Security	bcrypt	Industry-standard adaptive hashing for password storage
Environment Config	dotenv	Secure management of environment variables outside the codebase

10. Complete System Workflow

This section describes the end-to-end data flow from a business owner's initial registration to their listing appearing live on the correct country subdomain.

10.1 Business Owner Onboarding Flow

Step 1 Registration — Owner submits registration form. System creates an unverified user record and sends a verification email.

Step 2 Email Verification — Owner clicks the time-limited verification link. Account is marked as verified and login is permitted.

Step 3 Authentication — Owner logs in. System validates credentials, issues a signed JWT, and returns the token to the client.

Step 4 Business Profile — Owner submits business details. System creates a business record linked to user, country, city, and category with is_approved set to false.

Step 5 Media Upload — Owner uploads logo, images, video, and business card. Files are stored on Cloudflare R2 and URLs are saved in the relevant media tables.

Step 6 Services and Branches — Owner adds service listings and branch locations. Records are written to business_services and business_branches.

Step 7 Subscription Selection — Owner selects a plan. System initiates a Stripe Checkout session with the corresponding price ID.

Step 8 Payment Completion — Owner completes payment on Stripe's hosted page. Stripe fires a webhook event to the backend.

Step 9 Webhook Processing — Backend receives the Stripe webhook, verifies the signature, creates a subscriptions record with status 'active', and logs the payment in the payments table.

Step 10 Admin Notification — System sends an email to the admin notifying them of a new listing pending review.

Step 11 Admin Approval — Admin reviews the listing and approves it. System sets is_approved to true and sends the owner an approval confirmation email.

Step 12 Listing Published — Business is now publicly visible on the correct country subdomain with active status and full profile.

10.2 Public Directory Access Flow

Step 1 Subdomain Visit — Visitor navigates to a country URL such as korea.digitaldirectory.com.

Step 2 Country Resolution — Backend middleware extracts 'korea' from the host header, validates it against the countries table, and attaches country_id to the request context.

Step 3 Data Filtering — All queries for this request are scoped to the resolved country_id, returning only Korean businesses.

Step 4 Search Results — System returns approved, active-subscription businesses matching any applied search or filter parameters.

Step 5 **Profile View —** Visitor views a full business profile including media, services, branches, and contact details.

10.3 Subscription Expiry and Renewal Flow

Step 1 **Daily Expiry Check —** A scheduled background job runs each day and identifies subscriptions expiring within 7 days and within 1 day.

Step 2 **Reminder Emails —** NodeMailer sends automated reminder emails to affected business owners at both checkpoints.

Step 3 **Expiry Action —** On the expiry date, the subscription status is set to 'expired' and the business listing is automatically deactivated.

Step 4 **Owner Renewal —** Owner logs in, sees the renewal prompt, selects a plan, and completes the Stripe payment.

Step 5 **Webhook Confirmation —** Stripe webhook confirms the renewal payment. Backend creates a new subscriptions record, reactivates the listing, and sends a confirmation email.

11. Future Enhancements

The following features are planned for future development phases. They have been excluded from the current scope to maintain focus and delivery timeline but are architected for in the current database and system design.

- **Reviews and Ratings:** Public users can leave star ratings and written reviews on business profiles. A reviews table will extend the existing businesses entity.
- **Country-Level Admin Roles:** A dedicated Country Admin role that has management rights limited to their assigned country, without access to global settings.
- **Mobile Applications:** Native iOS and Android apps built on the same backend API, extending the platform to mobile users.
- **Business Owner Analytics Dashboard:** A self-service analytics view for business owners showing profile views, search appearances, and click-through rates.
- **SMS Notifications:** An additional notification channel alongside email, using an SMS gateway for subscription alerts and critical updates.
- **Multi-Language Support:** Interface and content localization supporting multiple languages per country subdomain.
- **Featured Listings and Promotions:** Premium placement options where businesses can pay to appear at the top of search results within their category or city.
- **Business Verification Badges:** An optional verification process where the platform validates the legitimacy of a business and displays a verified badge on the profile.

12. Conclusion

The Digital Directory platform is a carefully architected, centralized, and scalable multi-country business directory system. By leveraging a single backend instance with intelligent subdomain-based country routing, the platform achieves the rare combination of geographic flexibility and operational simplicity.

The database design is normalized and non-redundant, built to grow to any number of countries, cities, businesses, and subscribers without structural changes. The subscription system backed by Stripe ensures reliable revenue flow, while the notification layer and admin controls provide full operational visibility and management capability.

This SRS document represents the complete specification for the platform, covering all functional requirements, non-functional requirements, user journeys, database schema, and architectural decisions. It is intended to serve as the definitive reference for development, review, and client communication throughout the project lifecycle.

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